Strength of Materials Syllabus

MCE204 • Leonardo Gonzalez

Science and Engineering • Mechanical Engineering

1. Learning Objectives & Taxonomy

ID	Learning Objective	Bloom's Level
LO1	Understand fundamental concepts of stress, strain, and mechanical properties of materials.	Remember, Understand
LO2	Apply principles of axial loading, thermal stress, and analyze statically indeterminate structures.	Apply, Analyze
LO3	Analyze torsional stresses and deformations in shafts, and design for power transmission.	Analyze, Apply
LO4	Construct and interpret shear force and bending moment diagrams, and calculate bending and shear stresses in beams.	Apply, Analyze
LO5	Analyze combined loading scenarios and transform stresses using Mohr's Circle.	Analyze
LO6	Calculate beam deflections using integration, superposition, and energy methods.	Apply, Analyze
LO7	Evaluate column stability, predict buckling loads, and apply failure theories (yield criteria, fracture, fatigue) for material design.	Evaluate, Apply, Create

Overall Course Cognitive Depth: Understand, Apply, Analyze, Evaluate, Create

2. Weekly Instructional Plan

Wk	Subject and Activities	Outcomes
1	**Introduction to Strength of Materials: Stress, Strain, and Mechanical Properties of Materials** Face to Face Lectures, Syllabus • Discussion Forums	LO1
2	**Axial Loading: Normal Stress and Strain, Hooke's Law, and Deformations** Face to Face Lectures • Tutorial Sessions	LO2
3	**Thermal Stress, Strain Energy, and Statically Indeterminate Structures under Axial Loading** Study Questions, Face to Face Lectures	LO2
4	**Torsion: Torsional Shear Stress, Angle of Twist, and Power Transmission** Face to Face Lectures • Simulations • Assignments / Homework	LO3
5	**Bending: Shear Force and Bending Moment Diagrams** Face to Face Lectures • Recitation Sessions	LO4
6	**Bending Stress: Flexure Formula, and Composite Beams** Face to Face Lectures • Simulations	LO4
7	**Shear Stress in Beams: Shear Flow, and Thin-Walled Members** Face to Face Lectures • Lab Sessions • Midterm Exam	LO4
8	**Combined Loading and Stress Transformation: Normal and Shear Stresses on Inclined Planes** Face to Face Lectures • Tutorial Sessions	LO5
9	**Deflection of Beams: Integration Methods (Double Integration, Macaulay's Method)** Face to Face Lectures • Collaborative Study Sessions	LO6
10	**Deflection of Beams: Superposition Principle and Castigliano's Theorem** Face to Face Lectures • Individual Projects, Problem Sets	LO6
11	**Column Stability: Euler's Buckling Formula and Critical Load** Real-World Cases, Face to Face Lectures	LO7
12	**Intermediate Column Design: Secant Formula, Eccentric Loading, and Short/Intermediate Columns** Face to Face Lectures • Simulations	LO7
13	**Failure Theories: Yield Criteria (Tresca, Von Mises) and Fracture Mechanics Introduction** Face to Face Lectures • Group Projects	LO7
14	**Impact Loading and Fatigue: Stress Concentration, S-N Curves, and Endurance Limit** Real-World Cases, Face to Face Lectures	LO7

Wk	Subject and Activities	Outcomes
15	**Course Review and Advanced Topics (e.g., Introduction to Finite Element Analysis or Energy Methods Review)** Practice Exams • Collaborative Study Sessions • Problem Sets	—
16	**Final Exam** Final Exam	—

3. Skill Acquisitions by Category

Domain	Specific Skill	Related Wk
Hard	Foundational Terminology	1
	Unit Consistency	1
	Material Constitutive Relationships	1
	Force Equilibrium Analysis	2
	Stress-Strain Relationships	2
	Precise Deformation Calculations	2
	Methodical Problem Decomposition	3
	Solving Systems of Equations	3
	Analyzing Structures with Redundant Supports	3
	Quantifying Energy Absorption	3
	Calculating Shear Stresses (torsion)	4
	Calculating Deformations Due to Twisting Moments	4
	Material Selection Criteria for Torsional Applications	4
	Free-Body Diagramming	5
	Equilibrium Analysis	5
	Graphical Methods for Visualizing Internal Force Distributions	5
	Calculating Maximum Bending Stresses	6
	Locating the Neutral Axis	6
	Understanding Stress Variations Across Beam Cross-Sections	6
	Multi-Material Construction Analysis	6
	Applying General Shear Formula	7
	Understanding Parabolic Distribution of Shear Stress	7
	Analyzing Built-Up Sections	7
	Designing Against Shear Failure	7
	Identifying Critical Locations for Stress Concentrations	7, 14
	Superimposing Stress Components	8
	Understanding Principal Stresses and Maximum Shear Stress	8
	Applying Stress Transformation Equations	8

Domain	Specific Skill	Related Wk
	Predicting Failure in Complex Stress States	8, 13
	Derivation of Elastic Curve Equations	9
	Systematic Application of Boundary Conditions	9
	Integrating Load, Shear, and Moment Functions	9
	Determining Slopes and Deflections	9
	Applying Superposition Principle to Multi-Load Beams	10
	Utilizing Castigliano's Theorem	10
	Efficient Analysis of Complex Structures	10
	Calculating Critical Buckling Loads	11
	Understanding Limitations of Euler Formula	11
	Distinguishing Strength Failure and Stability Failure	11
	Selecting Appropriate Column Formulas	12
	Calculating Stresses Under Eccentric Loads	12
	Understanding Interaction of Axial Compression and Bending	12
	Choosing and Applying Appropriate Failure Criteria	13
	Calculating Equivalent Stresses (tresca, Von Mises)	13
	Understanding Factors Influencing Brittle Fracture	13
	Calculating Impact Stresses	14
	Interpreting S-N Diagrams	14
	Estimating Fatigue Life	14
	Designing Durable Components	14
	Exposure to Industry-Standard Simulation Techniques (FEA)	15
Soft	Practical Design Considerations	12
	Ensuring Safety and Reliability	13
Meta	Problem-Solving Methodologies	2, 5
	Integrated Understanding	3, 15
	Deeper Theoretical Insight (energy Principles)	10

Domain	Specific Skill	Related Wk
	Analytical Skills Refinement	15
	Integrating Diverse Knowledge Sets	15
	Reinforcing Conceptual Understanding	15
	Critical Thinking	15, 16
	Demonstrating Analytical Rigor	16
	Deep Understanding of Material and Structure Behavior	16

4. Proposed Micro-Credential Framework

Credential Name	Dimensions	Depth	Align.	Mapping
Structural Stress & Strain Analysis Specialist	Foundational Knowledge, Application	Proficiency	W1-3	#FoundationalTerminology #UnitConsistency #MaterialConstitutiveRelationships #ForceEquilibriumAnalysis #StressStrainRelationships #PreciseDeformationCalculations #MethodicalProblemDecomposition #SolvingSystemsOfEquations #AnalyzingStructuresWithRedundantSupports #QuantifyingEnergyAbsorption
Beam & Shaft Mechanics Professional	Foundational Knowledge, Application, Integration	Proficiency	W4-7	#CalculatingShearStresses(Torsion) #CalculatingDeformationsDueToTwistingMoments #MaterialSelectionCriteriaForTorsionalApplications #FreeBodyDiagramming #EquilibriumAnalysis #GraphicalMethodsForVisualizingInternalForceDistributions #CalculatingMaximumBendingStresses #LocatingTheNeutralAxis #UnderstandingStressVariationsAcrossBeamCrossSections #MultiMaterialConstructionAnalysis #ApplyingGeneralShearFormula #UnderstandingParabolicDistributionOfShearStress #AnalyzingBuiltUpSections #DesigningAgainstShearFailure

Credential Name	Dimensions	Depth	Align.	Mapping
Advanced Structural Analysis Expert	Application, Integration, Analysis	Mastery	W8-10	#SuperimposingStressComponents #UnderstandingPrincipalStressesAndMaximumShearStress #ApplyingStressTransformationEquations #PredictingFailureInComplexStressStates #DerivationOfElasticCurveEquations #SystematicApplicationOfBoundaryConditions #IntegratingLoadShearAndMomentFunctions #DeterminingSlopesAndDeflections #ApplyingSuperpositionPrincipleToMultiLoadBeams #UtilizingCastiglianosTheorem #EfficientAnalysisOfComplexStructures
Structural Integrity & Failure Prevention Engineer	Application, Integration, Evaluation, Caring	Mastery	W11-14	#CalculatingCriticalBucklingLoads #UnderstandingLimitationsOfEulerFormula #DistinguishingStrengthFailureAndStabilityFailure #SelectingAppropriateColumnFormulas #CalculatingStressesUnderEccentricLoads #UnderstandingInteractionOfAxialCompressionAndBending #ChoosingAndApplyingAppropriateFailureCriteria #CalculatingEquivalentStresses(TrescaVonMises) #UnderstandingFactorsInfluencingBrittleFracture #EnsuringSafetyAndReliability #CalculatingImpactStresses #InterpretingSN_Diagrams #EstimatingFatigueLife #DesigningDurableComponents

5. Assessment & Mapping

Activity Name	Week	Weight	Bloom's	Outcomes
Assignments	4	5%	Apply	LO3
Midterm Exam	7	20%	Apply, Analyze	LO1, LO2, LO3, LO4
Individual Projects	10	15%	Apply, Analyze	LO5, LO6
Problem Sets	10	5%	Apply, Analyze	LO5, LO6
Group Projects	13	25%	Apply, Analyze, Create	LO2, LO4, LO7
Final Exam	16	30%	Apply, Analyze, Evaluate, Create	LO1, LO2, LO3, LO4, LO5, LO6, LO7